Investigating Mechanisms of Hemodynamic Instability in Artificial Placenta Systems Using Lumped Parameter Models

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Background

- Conventional ventilation techniques may not be suitable for supporting extremely premature neonates with underdeveloped lungs [1].
- Artificial Placenta (AP) systems are being tested in animal models.
- The modeling in this research is based on preclinical data from systems being developed by Dr. Haller and collaborators at SickKids.

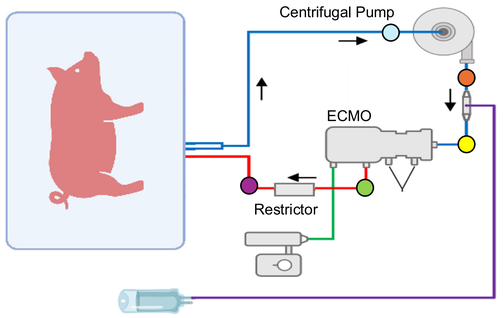
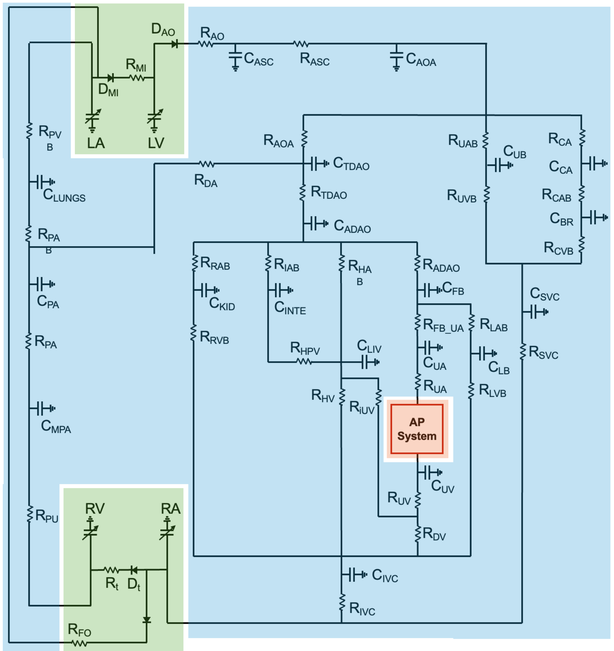


Figure 1: Experimental AP System Diagram

Methods



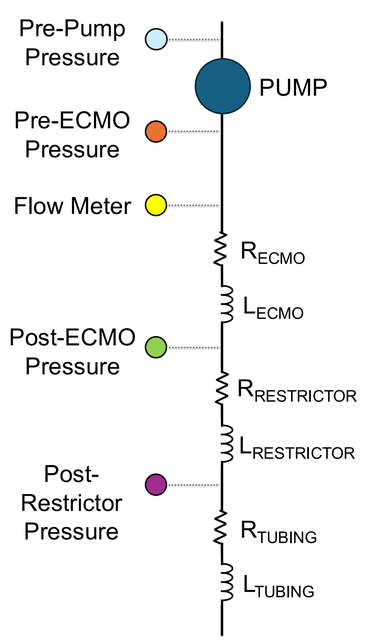


Figure 2: AP System with Fetal Model

1 Model Architecture and Scaling

LPM inspired by Yigit [2] and Pennati [3]; scaled from 3500 g to 1000 g. The AP system is based on the experimental setup at SickKids [4].

2 Physiological Mechanisms

Twelve physiological mechanisms added; the arterial baroreflex and the Frank-Starling mechanisms had the biggest impacts.

3 Parameter Tuning and Calibration

Most sensitive parameters (profile likelihood analysis) tuned with SLSQP optimization to minimize error in mean flow, pulse amplitude, and pressure drop vs. experimental data.

Objectives

Develop a computational Lumped Parameter Model (LPM) of the fetal circulation coupled to an artificial placenta system to investigate physiological mechanisms that may cause hemodynamic instability.

Results

The tuned LPM outputs were compared with experimental data from the Kühle et al. AP circuit [4] over a 30 s window.

Without mechanisms

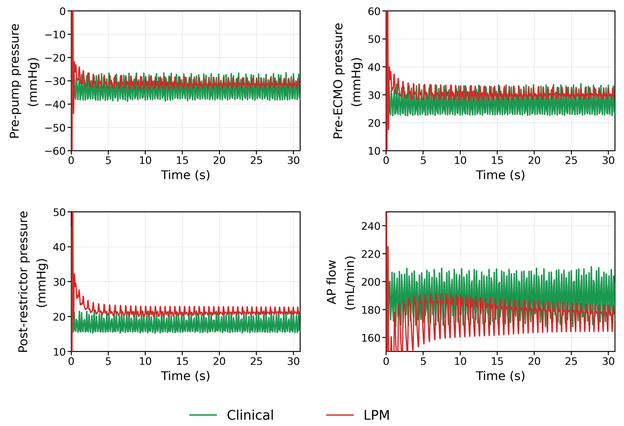


Figure 3: Model-generated Pressures and Flows of Baseline Model After Tuning

With mechanisms

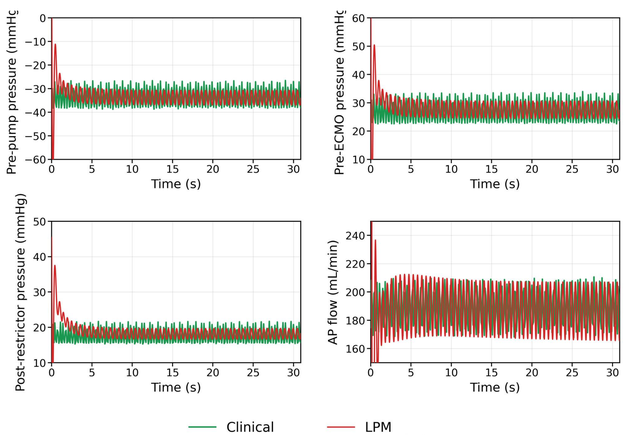
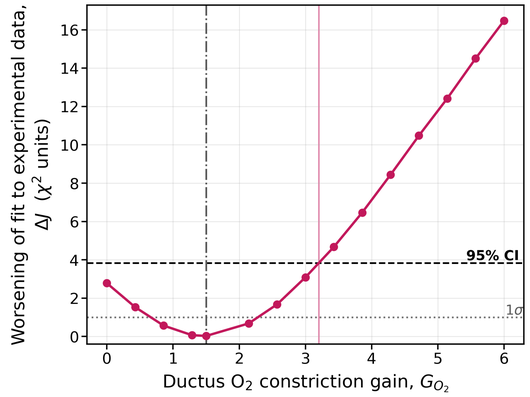
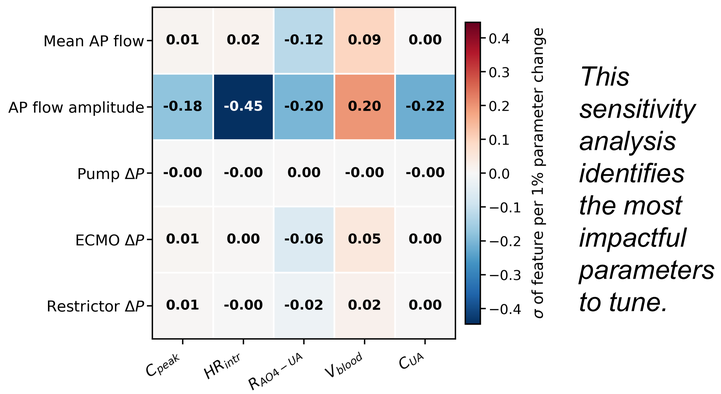


Figure 4: Model-generated Pressures and Flows with Mechanisms Added



This profile likelihood analysis provides insight into the practical identifiability of the parameters.

Figure 5: Profile Likelihood analysis for Ductus O_2 gain



This sensitivity analysis identifies the most impactful parameters to tune.

Figure 6: Sensitivity Analysis for LPM parameters

Key Findings

- **Flow in the AP circuit was the most sensitive output.**
- **Heart rate and blood volume were the most important parameters to tune.**

Conclusions

- We created and calibrated an LPM of the fetal circulation and AP system which reproduces experimental data.
- Future work will tune physiological mechanisms to identify potential causes of hemodynamic instability.

References

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SickKids